Cut elimination for G3c with the cut rule

The Open Logic Project contributors

openlogic:proof-theory/cut-elimination:local-snapshot-20261013;
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Editorial note. The counted statement is the author theorem labelled thm:cut-elim-G3c-largest (749-751); the proof span 753-771 is the author proof environment that immediately follows it.

Editorial note. External reference numbers render the unit-internal labels with hyphens in place of colons (pipeline number charset); the Source reference lines below the body give each label's exact source span.

Editorial scope. One original result of the Open Logic Project unit proof-theory/cut-elimination, counted once: cut elimination for G3c with the cut rule. Only the counted theorem and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Cut rank. The rank of a cut is the size of the cut formula; the maximal rank of a proof is the largest rank of its cuts.. The proof stratifies cuts by rank d and counts n the cuts of maximal rank (754-756).
- Maximal cut reduction. A topmost cut of maximal rank in a G3c proof can be replaced so that the end-sequent is unchanged and all cuts in the replaced part have strictly smaller rank; the unit proves this as its own lemma.. It performs the replacement step in the inner induction (758-762).
- Topmost cut. A cut is topmost when no cut of the same or greater rank occurs above it in the proof tree.. Picking a topmost cut of rank d ensures the replacement does not disturb other maximal cuts above it (758-759).

Source reference. Original lemma lem-max-cut-red-G3c: sources/olp-proof-theory-cut-elimination/pinned-original-source.tex, source lines 528–528. The author proof delegates the replacement of a topmost maximal cut to the unit's reduction lemma. The load-bearing work of that replacement is carried by the separately proved lemma of the same unit; the counted proof retains its own double induction in full.

Theorem 0.1. *Any !!proof π in* L*G3c* + Cut *can be transformed into a proof in* L*G3c.*

Proof. First we prove that all cuts of maximal rank in π can be removed. Let d be the maximal rank of cuts occuring in π, and let n be the number of cuts of rank d. The proof is by induction on n. If $n = 0$ there is nothing to prove. If $n > 0$, pick a topmost cut of rank d, and let π_1 be the sub-!!proof ending in it. By lem-max-cut-red-G3c, there is !!aproof π_1' of the same end-sequent with all cuts of rank $< d$. Replace π_1 by π_1' in π. This results in !!aproof containing $n - 1$ cuts of rank d, and the induction hypothesis applies.

Now the result follows by induction on d. If $d = 0$, all cuts are atomic, and by the previous result can all be removed. If $d > 0$, the previous result yields a proof in which all cuts of rank d have been replaced by cuts of rank $< d$. Since d was the maximum rank of cuts in π, we have proof in which the maximal rank of cuts is $< d$, and the induction hypothesis applies. □

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